

**Before the
U.S. Copyright Office
Library of Congress**

In the Matter of:

Software-Enabled Consumer Products Study:
Notice and Request for Public Comment

Docket No. 2015-6

COMMENTS OF THE ENTERTAINMENT SOFTWARE ASSOCIATION

The Entertainment Software Association (“ESA”) appreciates the opportunity to provide these Comments in response to the Copyright Office’s Notice of Inquiry (“NOI”) concerning software-enabled consumer products. *Software-Enabled Consumer Products Study: Notice and Request for Public Comment*, 80 Fed. Reg. 77,668 (Dec. 15, 2015). ESA is the U.S. trade association serving companies that publish computer and video games for video game consoles, handheld gaming devices, personal computers, and the Internet. It represents nearly all of the major gaming platform providers and major video game publishers in the U.S.¹

ESA welcomes the Office’s examination of copyright protection for embedded software. While this is a “highly specific” study focused solely on embedded software in everyday devices,² which therefore excludes software in general and video game platforms and game software in particular, ESA provides its perspectives to inform the Copyright Office about the ways in which software copyright protection has affected the video game industry.

¹ See <http://www.theesa.com/about-esa/members/> (listing ESA’s members).

² See 80 Fed. Reg. at 77,668 (“This is a highly specific study not intended to examine or address more general questions about software and copyright protection.”).

ESA believes that strong copyright protection for software (embedded and otherwise) has been a tremendous policy success that has enabled decades of innovation and creativity in product design and functionality. The video game industry in particular has been a leader in this innovation through the development of new services and business models that depend upon secure, trusted platforms enabled by software. More than thirty years ago, the federal courts confirmed that embedded software is subject to copyright protection on the same basis as other software. Ever since, that settled law has enabled innovation in the design and functionality of video game platforms and the games, online marketplaces and content delivery systems they support.

Given that record of success, ESA encourages the Office to approach this study with caution and a clear eye on basic copyright principles. In this study, the Office may hear suggestions for weakening software copyright protection or introducing supposed consumer protection regulations that have some superficial appeal. However, the current system of protection has enabled four decades of unprecedented technological progress. Whatever isolated issues might be raised concerning certain products with incidental embedded software should not be used as justification for a general weakening of software copyright protection or for regulation of the commercialization of copyrighted software in general.

Moreover, ESA has concern with the apparent assumption of the study that it might be appropriate to identify some nebulous category of software or “everyday products” for special treatment under the Copyright Act. Because of the range of products containing software, the range of software contained in such products, the range of distribution practices associated with such software, and the pace of change in all those characteristics, ESA believes that it would be

unwise to try to draw any such lines, and that any attempt to do so could have inadvertent consequences for software developers and consumers.

I. Introduction

As the NOI notes, products containing embedded software have become ubiquitous. This ubiquity has yielded tremendous benefits for consumers, who have access to exciting new classes of products; safer, “smarter,” more efficient and easier to use versions of older types of products; and new services and business models enabled by devices incorporating embedded software.

However, while the number, variety, and functionality of consumer products incorporating embedded software has grown substantially in recent years, the phenomenon of embedded software and the benefits it brings is not at all recent. The same things were beginning to happen in the 1970s as today’s legal framework for the copyright protection of computer software took shape. The drafters of the Copyright Act of 1976 and the members of the National Commission on New Technological Uses of Copyrighted Works (“CONTU”) were doubtless familiar with digital watches,³ electronic calculators,⁴ microwave ovens⁵ and other products incorporating embedded software that were revolutionary at their time. Some of those drafters or commission members may have driven a car with embedded software in its emission control system.⁶ And while gaming devices are probably not what the Office envisioned as “everyday products,” some of those drafters or commission members may even have played

³ See, e.g., <http://blog.onlineclock.net/earliest-digital-watches/>.

⁴ See, e.g., http://www.vintagecalculators.com/html/hand-held_calculators.html.

⁵ See, e.g., http://www.smecc.org/microwave_oven.htm.

⁶ In 1976, Volvo introduced a computer-controlled catalytic emission control system in its model 240 sold in California. John L. King and Kalle Lyytinen, *Automotive Informatics: Information Technology and Enterprise Transformation in the Automotive Industry*, at 7, *available at* <http://citeseer.ist.psu.edu/viewdoc/download;jsessionid=DA0E78E780ED411470CF150AB5E324A7?doi=10.1.1.136.2475&rep=rep1&type=pdf>.

games on a home console such as the Magnavox Odyssey (introduced in 1972)⁷ or Pong console (introduced in 1975)⁸ or on a handheld device such as Mattel Auto Race (introduced in 1976)⁹ or Mattel Football (introduced in 1977).¹⁰ By 1980, game software was available embedded in a wristwatch.¹¹

Since the enactment of the 1976 Act, copyright has played a critical role in promoting creativity and innovation in the successors to these products and the many other products that today incorporate embedded software. Early, seminal cases under the 1976 Act confirmed that software is protected by copyright even when embodied in read-only memory (“ROM”), as is the case for embedded software, and that the copyrights in such software are infringed when others incorporate that software into their products without authorization.¹² Industry has relied upon these principles for decades to justify the high costs of developing embedded software.¹³ Absent copyright protection, embedded software readily could be copied by others who might wish to introduce competitive products without expending the effort required to create their own original

⁷ See, e.g., http://americanhistory.si.edu/collections/search/object/nmah_1302004.

⁸ See, e.g., <https://www.atari.com/history/1972-1984-0>.

⁹ See, e.g., <http://www.handheldmuseum.com/Mattel/AutoRace.htm>.

¹⁰ See, e.g., <http://www.handheldmuseum.com/Mattel/FB.htm>.

¹¹ See, e.g., <http://www.digital-watch.com/DWL/1work/1251/> (describing the Casio GM-10 watch); <http://pocketcalculatorshow.com/nerdwatch/1980-casio-gm-10-game-watch/>.

¹² See, e.g., *Apple Computer, Inc. v. Franklin Computer Corp.*, 714 F. 2d 1240, 1249 (3d Cir. 1983) (computer program embedded in ROM of personal computer protectable and infringed by knockoff computer); *Williams Electronics, Inc. v. Artic Intern., Inc.*, 685 F. 2d 870 (3d Cir. 1982) (video game embedded in ROM of arcade game machine protectable and infringed by seller of competing components); *Tandy Corp. v. Personal Micro Computers, Inc.*, 524 F. Supp. 171 (N.D. Cal. 1981) (computer program embedded in ROM of personal computer protectable). The same result is now required by Article 10(1) of the Agreement on Trade-Related Aspects of Intellectual Property Rights, which provides that computer programs shall be protected as literary works under the Berne Convention “whether in source or object code.”

¹³ A typical new car has approximately 100 million lines of software code. Doug Newcomb, *The Next Big OS War Is in Your Dashboard*, *Wired* (Dec. 3, 2012), available at <http://www.wired.com/2012/12/automotive-os-war/all/>. Developing such software must represent a major part of the cost of developing the car. Software development certainly represents a major part of the cost of developing a new video game console.

works. The public has benefitted from the outpouring of innovation that has resulted from copyright protection for both embedded and other software.

Software copyright protection has been especially important to the growth of the video game industry. For 40 years, video game companies have been on the cutting edge of creativity and innovation and, in 2015, the industry generated U.S. revenues of \$23.5 billion from sales of hardware, game software, other content, accessories and services.¹⁴ During that time, the nature of the software used in gaming devices has evolved considerably. Many early devices, such as the Pong console and various handheld devices and game watches, played only a limited repertoire of games embodied in software that was embedded in the device when it shipped. However, in 1972 the Magnavox Odyssey used removable game cards to provide a more varied game experience, and in 1976, Fairchild introduced the Channel F, a console in which games embodied in programmable read only memory cartridges were executed on a microprocessor.¹⁵ The industry quickly migrated to a model in which system software embedded in a device played game software distributed on removable media such as cartridges and eventually optical discs, and in due course by means of download. With the passage of time, the software that operates video game consoles and handheld devices has become increasingly extensive and sophisticated, and today, many game devices rely less on embedded software and more on updateable operating systems and cloud services. The software also performs more activities and provides more features and services than just gaming.

¹⁴ ESA, U.S. Video Game Industry Generates \$23.5 Billion in Revenue for 2015, *available at* <http://www.theesa.com/article/u-s-video-game-industry-generates-23-5-billion-in-revenue-for-2015/>; *see also* David Greenspan, S. Gregory Boyd and Jas Purewal, Video Games and IP: A Global Perspective, WIPO Magazine (Apr. 2014) (video games are “the fastest growing sector of the entertainment industry and an important driver of economic growth, creating millions of jobs, generating much-needed tax revenues and offering exciting opportunities for talented creators and engineers from all corners of the globe”), *available at* http://www.wipo.int/wipo_magazine/en/2014/02/article_0002.html.

¹⁵ Benj Edwards, The Untold Story of the Invention of the Game Cartridge, Fast Company, *available at* <http://www.fastcompany.com/3040889/the-untold-story-of-the-invention-of-the-game-cartridge>.

Today, more than 150 million Americans play video games, and 42 percent of them play video games regularly, or at least three hours per week.¹⁶ Each of the three major home consoles works in tandem with an integrated, console-specific online network.¹⁷ Those networks provide a wide array of functions in support of game play, including enabling cooperative and competitive multiplayer game play, tracking and accumulation of scores and achievements across devices, downloading of games, announcements for new games, expansion packs and updates, tournaments, social and community features, and customization of avatars. Game consoles also provide social media features relevant to gaming, such as the ability to capture and share excerpts of game play and stream gameplay over streaming networks like Twitch.tv. These features are not merely adjuncts to the console (and the embedded software therein), but are now viewed as integral to the overall console and game experience.

These networks also provide access to non-game entertainment services, including films, television shows and music. While the specific offerings vary from platform to platform, the major gaming platforms collectively provide access to dozens of apps, including video streaming from services such as Amazon, HBO GO, Hulu, Netflix and YouTube; sports from the “big four” professional leagues and Watch ESPN, and music from Microsoft’s Groove music service, Spotify, Pandora and iHeartRadio.¹⁸ Other options on some gaming devices include rental or purchase of movies and television shows and communications through realtime audio chat and Skype.

¹⁶ ESA Essential Facts about the Computer and Video Game Industry, at 2 (Apr. 2015), *available at* <http://www.theesa.com/wp-content/uploads/2015/04/ESA-Essential-Facts-2015.pdf>.

¹⁷ See, e.g., Xbox LIVE, <http://www.xbox.com/en-US/live>; Nintendo Wii U eShop, <http://www.nintendo.com/wiiu/eshop>; PlayStation Entertainment, <http://www.playstation.com/en-us/explore/playstationnetwork/apps-and-entertainment/>.

¹⁸ For more complete lists of available apps see <http://support.xbox.com/en-US/xbox-live-status>; <https://www.playstation.com/en-us/explore/playstationnetwork/entertainment/> and <http://www.nintendo.com/eshop/apps>.

These features are implemented with layers of software residing on the game console, at least some of which would probably be considered embedded in the console, as well as software powering the various services. Copyright law is largely responsible for providing necessary incentives for both game and game device makers – ensuring they have the ability to expand features and access and still protect their innovation. Thus, the video game industry and its innovations to provide consumers with exciting new content – and new ways to interact with content – depend critically upon copyright protection for the software embedded in game consoles and handheld gaming devices (as well as for the various other copyrighted works involved). These innovations have also been possible because the government has wisely decided to take a light hand in regulating technological innovation.

Accordingly, ESA believes that the ubiquity of products containing embedded software should not, in general, be viewed as a problem for copyright law that might require new regulatory intervention, but instead as evidence of the success of copyright law at achieving its intended purposes. Such embedded software fits well into the current copyright scheme in a way that is indistinguishable from other software. Outside the software and video game industries, it is possible that the Office may hear complaints concerning commercial practices involving other kinds of products controlled by embedded software. However, ESA urges the Office to view anecdotal reports of such issues with due skepticism, and to keep solidly in view copyright's important role of promoting creativity and innovation in the development of software. The Office should exercise caution in weighing *de minimis* or hypothetical consumer-protection against the necessary role of strong copyright protection in ensuring creativity and innovation.

II. The Office's Specific Areas of Inquiry

1. The provisions of the copyright law that are implicated by the ubiquity of copyrighted software in everyday products

Software, whether embedded in everyday products or not, implicates all the generally-applicable provisions of the Copyright Act, as well as specialized provisions in Sections 109(b), 109(e) and 117. Importantly, owners of software copyrights enjoy all the exclusive rights specified in Section 106(1) through (5) and the full range of remedies for infringement specified in Chapter 5. As described above, these provisions provide the intellectual property foundation for the software and video game industries, and their protections are important whether software is designed to be used on a general-purpose computer or embedded in a video game console, handheld gaming device or other specialized product.¹⁹

Software is also subject to the generally-applicable limitations on the scope of copyright protection (as well as the specialized provisions of Section 117). These include the idea-expression dichotomy and fair use, which have traditionally been the mechanisms by which copyright law ensures that its important protections do not inadvertently inhibit creativity, innovation and expression. *See, e.g., Golan v. Holder*, 132 S. Ct. 873, 889-91 (2012); *Eldred v. Ashcroft*, 537 U.S. 186, 219-20 (2003). As the NOI observes, these limitations have proven flexible enough to accommodate innovation through development of products interoperable with those containing embedded software. *See Sony Computer Entertainment v. Connectix Corp.*, 203

¹⁹ While we understand that the office wishes to address circumvention of access and copy controls in its separate study of Section 1201, we feel compelled to note here the importance of Chapter 12's protections against circumvention. Video game consoles are at the center of an intellectual property ecosystem that makes copyrighted content easily and legally accessible, to the benefit of creators, distributors and the gaming public. Consoles include technological measures – principally in the form of embedded software – that protect not only the integrity of the console code, but the copyrighted works that run on and are available through the consoles. They also interact with server-based technological protection measures implemented in software. These measures provide incentives to create and disseminate video games and other content for consoles, and prevent individuals from making unauthorized copies of this content, distributing it to others or using the console to play pirated copies of such works. Protection of all this software against circumvention is thus critical to the video game ecosystem.

F. 3d 596, 602-08 (9th Cir. 2000) (disassembly of embedded software in game console, as a necessary step in examination of unprotected elements, was fair use); *Sega Enterprises Ltd. v. Accolade, Inc.*, 977 F. 2d 1510, 1520-28 (9th Cir. 1992) (same). Related limitations such as the merger and scènes à faire doctrines noted in the NOI play a role in enabling creativity and innovation in software development.²⁰ Software is also subject to Section 109’s first sale doctrine.²¹ With one exception involving the first sale doctrine,²² copyright law has not previously distinguished between software embedded in everyday products and other types of software. ESA is not convinced that such a distinction is warranted, or that such a distinction can be drawn in an administrable manner.

²⁰ See *Computer Associates Intern., Inc. v. Altai, Inc.*, 982 F. 2d 693, 708-10 (2d Cir. 1992) (merger doctrine permits copying of certain expression dictated by efficiency, and scènes à faire doctrine permits copying of certain expression constituting a standard technique); see also *Oracle America, Inc. v. Google Inc.*, 750 F. 3d 1339, 1359-62, 1363-64 (Fed. Cir. 2014) (discussing application of the merger and scènes à faire doctrines to software, but finding that copying of the particular software at issue was not proven to be permitted by those doctrines).

²¹ As the Office knows well, the first sale doctrine generally permits the owners of lawful copies of works to part with those copies. It “does not permit the distribution of a work through digital transmission where copies are created, because the reproduction right is implicated.” U.S. Dep’t of Commerce, White Paper on Remixes, First Sale, and Statutory Damages, at 35 (2016) [hereinafter “White Paper”], available at <https://www.ntia.doc.gov/report/2016/white-paper-remixes-first-sale-and-statutory-damages>. The Office long ago concluded that the first sale doctrine should not be extended to cover digital transmissions because of the differences between the transfer of a physical copy and an online transmission. U.S. Copyright Office, Report of the Register of Copyrights Pursuant to § 104 of the Digital Millennium Copyright Act, at 97-101 (2001), available at <http://www.copyright.gov/reports/studies/dmca/sec-104-report-vol-1.pdf>. The Commerce Department’s Internet Policy Task Force recently agreed that it could not “at this time recommend extending the first sale doctrine to apply to digital transmissions of copyrighted works.” White Paper at 58. ESA agrees with this conclusion.

²² An exception to the first sale doctrine provides so-called “rental rights” for most software, allowing the copyright owners of software to prevent rental by owners of copies. 17 U.S.C. § 109(b). The Act provides an exception to that exception for certain embedded software and video game console software. 17 U.S.C. § 109(b)(1)(B). Thus, owners of copies of those kinds of software are permitted to rent their copies, while owners of copies of other kinds of software are not.

2. Whether, and to what extent, the design, distribution, and legitimate uses of products are being enabled and/or frustrated by the application of existing copyright law to software in everyday products

As described above, copyright protection promotes – and is necessary to enable – creativity and innovation in the development of software and products containing software.²³ As some of the cases cited above (and others) show, in the absence of strong copyright protection for software, potential competitors can and will discern and copy embedded software, because that is an easier path to getting to market with a software-enabled product than creating original software.²⁴ As a result, copyright infringement is a concern for embedded software as well as for software intended to be used on a general-purpose computer. The determinations during the 1980s that software copyright protection extends to software in executable form, even when embodied in read only memory embedded in a consumer product, have thus been critical to the ubiquity of embedded software that the Office noted in the NOI. Far from frustrating the legitimate uses of products with embedded software, strong copyright protection has been essential to incentivize the very creation of such products.

3. Whether, and to what extent, innovative services are being enabled and/or frustrated by the application of existing copyright law to software in everyday products

Copyright protection for software, including embedded software, also enables innovative services. As ESA’s comments in the Office’s recently-completed anticircumvention rulemaking

²³ One way copyright protection achieves this result is by enabling licensing, which creates flexible relationships between product providers and their customers, which may include consumer benefits such as the provision of patches and other updates to improve functionality and fix security vulnerabilities, as well as new versions of software.

²⁴ *E.g., Apple v. Franklin*, 714 F. 2d at 1249 (copying of personal computer ROM software); *E.F. Johnson Co. v. Uniden Corp. of Am.*, 623 F. Supp. 1485, 1503 (D. Minn. 1985) (copying of certain elements of software embedded in radio).

describe in further detail,²⁵ video game consoles are the center of an intellectual property ecosystem that makes both video games and related items and other types of content such as music, movies and television programming easily and legally accessible, to the benefit of creators, distributors and the gaming public. The video game industry also provides robust, interconnected online experiences that leverage users' Internet connectivity to supplement game play. These immersive gaming experiences include multiplayer game play, chat communications, sharing of user-generated content, leaderboards, achievement markers, access to updated data for sports games, downloadable content, opportunities for users to create virtual worlds, and more realistic game experiences enabled by server-based game calculations. These services typically are offered separately from the purchase of the game console or any particular game itself. All of these services are anchored by embedded software in game consoles and handheld gaming devices that provides a user-friendly and secure platform for these services. Absent copyright protection for the embedded software to provide a secure, trusted platform for these services, copyright owners would be legitimately concerned that their content could easily be infringed if made available through such services. Weakening copyright protection for that embedded software could thus threaten services and business models that depend upon having a trusted platform for the dissemination of content.

4. Whether, and to what extent, legitimate interests or business models for copyright owners and users could be undermined or improved by changes to the copyright law in this area

ESA believes that current copyright law as applied to software (both embedded in everyday products and otherwise) is, in general, working reasonably well to enable innovation

²⁵ ESA Comment on Proposed Class 19 in Docket No. 2014-07 (Mar. 27, 2015), *available at* http://copyright.gov/1201/2015/comments-032715/class%2019/Entertainment_Software_Association_Class19_1201_2014.pdf; ESA Comment on Proposed Class 23 in Docket No. 2014-07 (Mar. 27, 2015), *available at* http://copyright.gov/1201/2015/comments-032715/class%2023/Entertainment_Software_Association_Class23_1201_2014.pdf.

and creativity not only as software-enabled products themselves, but as to products and services that depend upon such products. After decades of strong copyright protection for software that have seen more technological innovation than any similar period in human history, it is clear that the law does not need to change to enable new products, services or business models. The Office should take care to ensure that any proposed changes to copyright law do not jeopardize the benefits the public has realized from the incentives to creativity and innovation provided by strong copyright protection and the ability of platform providers to create secure, legally-protected environments for the dissemination of copyrighted content.²⁶

5. Key issues in how the copyright law intersects with other areas of law in establishing how products that rely on software to function can be lawfully used

Software is commonly licensed, including sometimes, embedded software. However, a competitive consumer marketplace provides ample incentives for software providers that wish to license their software, rather than selling copies of it, to adopt appropriate consumer licensing practices. It also has never been the role of copyright law to regulate consumer contracts. Now is not the time to begin using copyright law as a vehicle for imposing supposed consumer-protection requirements, nor should the Office advocate any government intervention into software licensing markets.

III. Conclusion

ESA welcomes the Office's examination of copyright protection for embedded software and believes that, when the record is closed, it will be clear that strong copyright protection for

²⁶ To the extent that this inquiry is based on concerns about the role of embedded software as to issues such as cellphone unlocking, access to motor vehicle software for maintenance, and medical device testing (issues that arose in the last triennial rulemaking), ESA understands that such concerns should be addressed within the context of the Office's Section 1201 study. See NOI, 80 Fed. Reg. at 77,669 ("[t]his study is not the proper forum for issues arising under section 1201 of the Copyright Act"). In any event, even the software implicated by such concerns deserves the full protection of copyright law.

software has been a tremendous policy success. While the Office may hear complaints of isolated issues or proposals for supposed consumer protection regulations, now is not the time to question basic principles of software copyright protection that have enabled decades of creativity and innovation and produced the ubiquitous presence of software in our daily lives. The Office should proceed cautiously to ensure that copyright continues to perform its role as an enabler of creativity and innovation for the next 40 years as much as it has for the last 40 years.

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